



# Scale Footprint

Derive a footprint for a new running-surface length from a reference: accordion in X, or keep the taper -- arcs stay tangent arcs that report their radius.

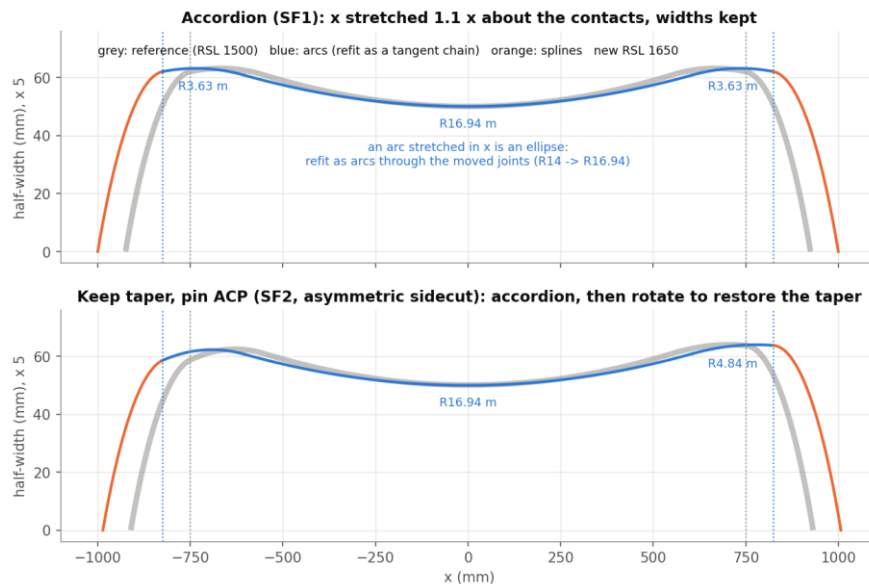


# What it does

A ski family shares one sidecut at several lengths. Scale Footprint moves the tip and tail to the new contacts and stretches the sidecut between them -- keeping arcs as arcs and, if asked, the reference taper.

## Use it when

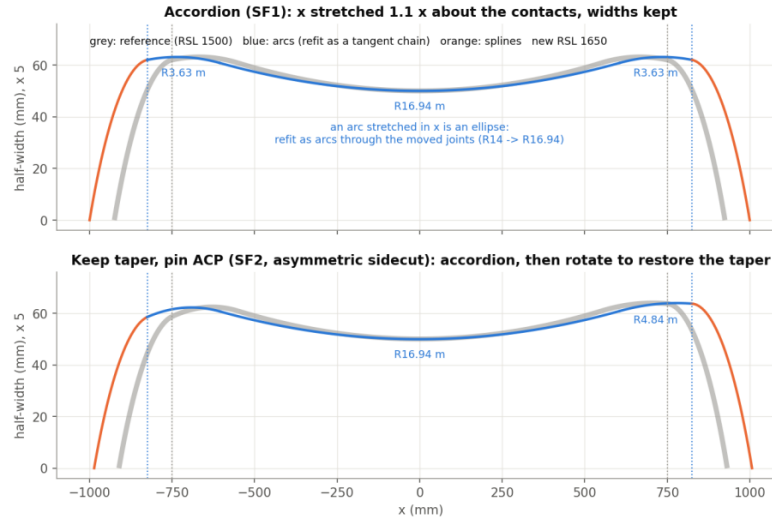
- Grow or shrink a ski family by length
- Keep a taper angle across lengths
- Keep radius callouts on the scaled outline



SF1 (accordion) and SF2 (keep taper, pin ACP): reference grey, arcs blue, splines orange.



# Arcs stay arcs



## Stretch = ellipse

A circle stretched in X is no longer a circle.

## Refit as a chain

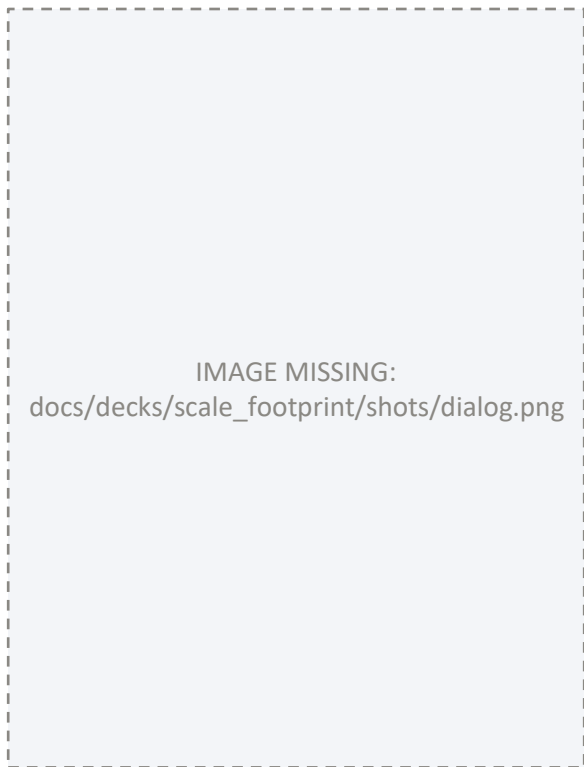
Arcs through the moved joints, each starting tangent to the last -- tangent by construction (R14 -> R16.94 m at x 1.1).

## Splines exact

A spline sidecut is stretched by its control points; knots kept (within 0.001 mm).



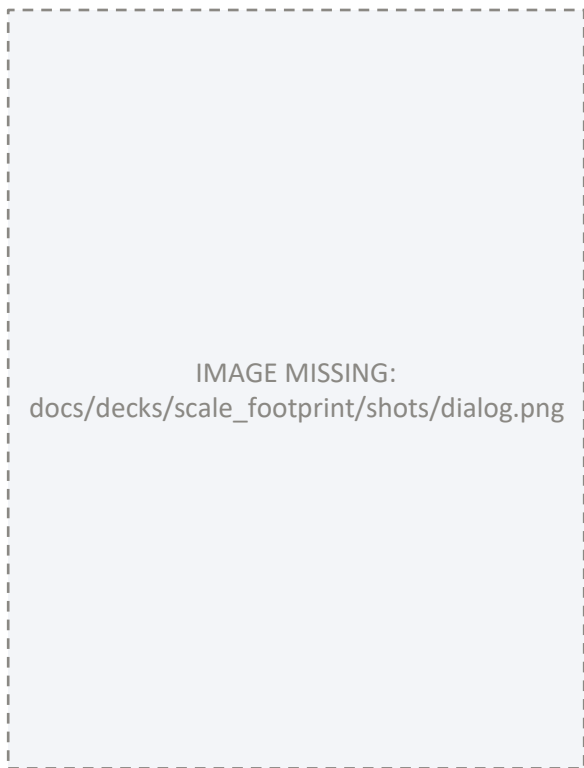
## The dialog (1/2)



| # | Parameter                 | What it means                                              |
|---|---------------------------|------------------------------------------------------------|
| 1 | Reference footprint edges | The whole reference, +Y and -Y.                            |
| 2 | Reference RSL line        | FCP = its lower-X end.                                     |
| 3 | New RSL line              | The new contacts.                                          |
| 4 | Symmetry mode             | Symmetric (mirror +Y) / Asymmetric (each side on its own). |
| 5 | Scale mode                | Accordion (X only) / Keep taper angle.                     |
| 6 | Pin location              | Keep taper: hold the ACP width or the MRS width.           |
| 7 | Spline options            | Degree, Strict arcs, tolerance, maximum control points.    |



## The dialog (2/2)



| #  | Parameter                      | What it means                                                |
|----|--------------------------------|--------------------------------------------------------------|
| 8  | Specify target width           | Target waist width (full).                                   |
| 9  | -Y Scale mode                  | Asymmetric: the -Y side's own settings.                      |
| 10 | Enforce tangency at tip (FCP)  | Move one tip control point so the tip joins tangentially.    |
| 11 | Enforce tangency at tail (ACP) | Same at the tail.                                            |
| 12 | Keep reference curves          | Keep the input.                                              |
| 13 | Debug                          | Colour curves by type; skip wire merge; print the arc chain. |



## Examples (Footprint tests)

### **SF1: accordion S14 -> RSL 1650**

Contacts +-825 at y 61.999; tip end -924.567  
-> -999.567; waist (0, 50) kept; -Y the mirror.

### **SF2: keep taper, pin ACP**

Asymmetric sidecut: taper -0.0662 deg kept  
(accordion alone: -0.0602); ACP width 63.666  
kept.

### **SF5: spline sidecut**

Output = the exact X-scaled reference spline,  
knots kept, within 0.001 mm.



# Outputs and messages

## Published for Extract variables

| Key                                | What it holds                                                |
|------------------------------------|--------------------------------------------------------------|
| <b>positiveSide / negativeSide</b> | The scaled +Y and -Y wires (open; never closed into a loop). |
| <b>lengthScale</b>                 | New RSL / reference RSL.                                     |

## Messages

|              | Message / what to do                                              |
|--------------|-------------------------------------------------------------------|
| <b>Error</b> | Asymmetric with no -Y data<br>Select both sides of the reference. |



## Tips and limits



### Pin choice

Pin ACP keeps the tail width; pin MRS keeps the mid-stance width.



### World axes

Works in world XY with the FCP at the lower-X end of each RSL line.



### No Scale radius

The retarget-radius mode was removed 2026-09-25 (it missed its target).





## Where it lives

| <b>Document</b>  | footprint (52724f3a857fa52d3ecceb77)                                                        |
|------------------|---------------------------------------------------------------------------------------------|
| <b>Tab</b>       | scaleFootprint (+ fpt_analyze, arcFit)                                                      |
| <b>Tests</b>     | Footprint tests studio (build_ / check_footprint_tests.py): SF1, SF2, SF5                   |
| <b>Explainer</b> | docs/explainers/footprint/footprint_explained.md, section 2.3                               |
| <b>Related</b>   | Integrate footprint, Analyze footprint, Scale Footprint, Arc fit, Generate Footprint Points |